Thesis Notes

Nathaniel Kenschaft

May 29, 2026

Contents

0.1 TODO

1. ~~Read [Far06]~~
2. ~~Read [CG13]~~
3. Read [DLS24]
4. Ok so there's no continuous section in [DLS24], but what is the Schwarz genus? I.e. how many cases are there?
5. Is it a good idea to just canonicalize via $\mathbb{S}_d^n$? Why or why not?
6. What can we learn about $X = \left\{A \in \mathbb{R}^{d\times n} : \operatorname{rank}(A) = d\right\}$ and $X/O(d)$ or $X/SO(d)$?
7. Is a canonicalization up to homotopy sufficient?
8. Use restrictions to bound sectional category on original space
9. Consider $\mathrm{SO}(d)$ acting on $\mathbb{R}^{d\times n}$
10. Understand proof that there is no canonicalization of $(\mathbb{R}^{d\times n}, \mathrm{O}(d))$ and $(\mathbb{R}^{d\times n}, \mathrm{SO}(d))$.
11. Consider what group acts on vectorized images and consider Schwarz genus
12. What if we want a *differentiable* section?

0.2 Notation

- I use $A \cong B$ to denote that there exists an isomorphism between A and B in whatever the appropriate category is. For instance if X and Y are topological spaces then $X \cong Y$ means X and Y are homeomorphic, or if G and H are groups then $G \cong H$ means G and H are isomorphic as groups.
- For a homotopy $h : X \times [0,1] \to Y$, I may also write $h_t : X \to Y$ with $h_t(x) := h(x,t)$.
- For any set X, I write c_x to mean the constant function at $x \in X$.
- Given a path $\gamma : [0,1] \to X$, I write $\overline{\gamma}$ to mean the reversed path $\overline{\gamma}(t) := \gamma(1-t)$.
- For any two paths $\alpha, \beta : [0,1] \to X$ with $\alpha(1) = \beta(0)$, I write $\beta * \alpha$ to indicate their concatenation:
$$(\beta * \alpha)(t) = \begin{cases} \alpha(2t) & \text{if } 0 \le t \le \frac{1}{2} \\ \beta(2t-1) & \text{if } \frac{1}{2} \le t \le 1 \end{cases}.$$
The exact paramerization is unimportant, but if it ever matters this is what I mean.
- Unless specified otherwise, for any group G I will write $e \in G$ to be the identity in G. If there is the potential for confusion between different groups, I will use subscripts to differentiate (e.g. e_G is the identity in G, e_H is the identity in H).
- I write $\mathbb{R}^{d\times n}_{\operatorname{rank} k}$ to mean the set $\left\{A \in \mathbb{R}^{d\times n} : \operatorname{rank}(A) = k\right\}$ and set $\mathbb{R}^{d\times n}_{*} = \mathbb{R}^{d\times n}_{\operatorname{rank} d}$.

Chapter 1

Background

1.1 Background from [Far06]

Definition 1.1. *For a topological space X and positive integer n,*

$$F(X,n) = \{(x_1, ..., x_n) \in X^n : x_i \neq x_j \text{ for } i \neq j\}.$$

Intuitively, this gives the configuration space of n particles moving through X with no collisions.

Definition 1.2. *For fixed $a = (a_1, ..., a_m) \in \mathbb{R}^m_+$, we define the variety*

$$M(a) = \left\{ (z_1, ..., z_m) \in (S^2)^m : \sum_{i=1}^m a_i z_i = 0 \right\} / SO(3)$$

where $SO(3)$ acts coordinate-wise on $(S^2)^m$. This variety describes all polygons in $\mathbb{R}^3$ with sidelengths $a_1, ..., a_m$.

Definition 1.3. *Let X be a path-connected topological space. Then $PX = C([0,1]; X)$ is the space of all paths $\gamma : [0,1] \to X$ equipped with the compact-open topology.*

Recall that the compact-open topology has a subbase given by sets of the form

$$V(K,U) := \{f \in C(X,Y) : f[K] \subseteq U\}$$

*for $K \subseteq [0,1]$ compact and $U \subseteq X$ open [**wikipedia:compact-open**].*

Definition 1.4. *For spaces E and B, a map $p : E \to B$ has the* homotopy lifting property (HLP) *for a space X if for every homotopy $h_t : X \to B$ and every lift $\widetilde{h_0} : X \to E$ of h_0 ($h_0 = p \circ \widetilde{h_0}$), there exists a homotopy $\widetilde{h_t} : X \to E$ that lifts h_t ($h_t = p \circ \widetilde{h_t}$).*

Definition 1.5. *For spaces E and B, a map $p : E \to B$ is a* (Hurewicz) fibration *if p satisfies the homotopy lifting property for all topological spaces [Wik26b].*

Remark 1.1. *The map $\pi : PX \to X \times X$ given by $\pi(\gamma) = (\gamma(0), \gamma(1))$ is a fibration.*

Proof. Let Y be any topological space and let $h_t : Y \to X \times X$ be a homotopy with some $\widetilde{h_0} : Y \to PX$ such that $h_0 = \pi \circ \widetilde{h_0}$. Then we simply define

$$\widetilde{h_t}(y)(s) = \begin{cases} (h_{t(1-3s)}(y))_1 & \text{if } 0 \leq s \leq \frac{1}{3} \\ \widetilde{h_0}(y)(3(s - \frac{1}{3})) & \text{if } \frac{1}{3} \leq s \leq \frac{2}{3} \\ (h_{t(3s-2)}(y))_2 & \text{if } \frac{2}{3} \leq s \leq 1 \end{cases}.$$

Intuitively, this simply defines a path from $(h_t(y))_1$ to $(h_t(y))_2$ by following:

1. $(h_\bullet(y))_1$ from $(h_t(y))_1$ to $(h_0(y))_1$,
2. $\widetilde{h_0}$ from $(h_0(y))_1 = \widetilde{h_0}(y)(0)$ ($h_0 = \pi \circ \widetilde{h_0}$) to $\widetilde{h_0}(y)(1)$,
3. and $(h_\bullet(y))_2$ from $\widetilde{h_0}(y)(1) = (h_0(y))_2$ ($\pi \circ \widetilde{h_0} = h_0$) to $(h_t(y))_2$.

By construction $\widetilde{h_t}$ is continuous, so we conclude that π is a fibration. $\square$

Definition 1.6. *A* motion planning algorithm *is a section* $s : X \times X \to PX$ *of* π*, i.e.* $\pi \circ s = \mathrm{id}_{X\times X}$.

Lemma 1.1. *A continuous motion planning algorithm in* X *exists if and only if* X *is contractible.*

Proof. First assume that there exists a continuous motion planning algorithm, i.e. a continuous section $s : X \times X \to PX$ of π, and fix some $x_0 \in X$. Then for every $x \in X$, define $\gamma_x = s(x_0, x) \in PX$. Now define a homotopy $h_t : X \to X$ via $h_t(x) = \gamma_x(t)$. Observe that $h_0(x) = \gamma_x(0) = x_0$ and $h_1(x) = \gamma_x(1) = x$ so $h_0 = c_{x_0}$ and $h_1 = \mathrm{id}_X$. Since s is continuous, this means that h is a homotopy from c_{x_0} to id_X and therefore X is contractible.

Now instead assume that X is contractible and let $h_t : X \to X$ be a homotopy exhibiting this ($h_0 = c_{x_0}$ for some $x_0 \in X$ and $h_1 = \mathrm{id}_X$). Then for each $x \in X$, define $\gamma_x \in PX$ via $\gamma_x(t) = h_t(x)$ and define $s : X \times X \to PX$ by $s(x, y) = \gamma_y * \overline{\gamma_x}$. Then s is a section since $\pi(s(x, y)) = ((\gamma_y * \overline{\gamma_x})(0), (\gamma_y * \overline{\gamma_x})(1)) = (x, y)$, and s is continuous by construction. $\square$

Definition 1.7. *A topological space* X *is a* Euclidean Neighborhood Retract (ENR) *if there exists an embedding* $i : X \to \mathbb{R}^n$ *for some* n *with an open neighborhood* U *of* $i[X]$ *such that* U *retracts onto* $i[X]$.

Definition 1.8. *A motion planning algorithm* $s : X \times X \to PX$ *is* tame *if there are finitely many disjoint sets* $F_i \subseteq X \times X$ *such that* $X \times X = \cup_i F_i$*,* $s|_{F_i}$ *is continuous, and* F_i *is an ENR.*

Definition 1.9. *The* topological complexity *of a tame motion planning algorithm is the minimum number of sets* F_i *one can take in definition 1.8.*

Definition 1.10. *For a path-connected space* X*, the topological complexity* $\mathbf{TC}_1(X)$ *is the minimum topological complexity of tame motion planning algorithms in* X*. We set* $\mathbf{TC}_1(X) = \infty$ *if* X *admits no tame motion planning algorithms.*

Remark 1.2. $\mathbf{TC}_1(X) = 1$ *if and only if* X *is a contractible ENR.*

Proof. This trivially follows from lemma 1.1 and definitions 1.7 and 1.8. $\square$

Definition 1.11. *The* Schwarz genus *or* sectional category *of a fibration* $p : E \to B$*, denoted* $\mathrm{secat}(p)$ *is the minimum number* k *of open sets* $U_1, ..., U_k$ *covering* B *such that each* U_j *admits a continuous section* $s_j : U_j \to E$ *of* p.

Remark 1.3. *The Schwarz genus of a fibration* $p : E \to B$ *is 1 if and only if* p *admits a continuous section* $s : B \to E$.

Definition 1.12. *For a path-connected pointed space* (X, x_0)*, we define*

$$P_0X := \{\gamma \in C([0, 1]; X) : \gamma(0) = x_0\}.$$

Definition 1.13. *For a path-connected pointed space* (X, x_0)*, the* Lusternik-Schnirelmann category of X*, denoted* $\mathrm{cat}(X)$*, is the Schwarz genus of the fibration* $\pi_0 : P_0X \to X, \gamma \mapsto \gamma(1)$.

Definition 1.14. *For a path-connected space X, the topological complextiy $\mathbf{TC}_2(X)$ is the Schwarz genus of the fibration $\pi : PX \to X \times X$.*

Remark 1.4. $\mathbf{TC}_2(X) = 1$ *if and only if X is contractible.*

Proof. This is a direct corollary of lemma 1.1. $\square$

Lemma 1.2. *Let $p : E \to B$ be a fibration, and let $B' \subseteq B$ with $E' := p^{-1}[B']$ and $p' = p|_{E'} : E' \to B'$. Then* $\operatorname{secat}(p') \leq \operatorname{secat}(p)$.

Proof. Assume the $\operatorname{secat}(p) < \infty$ and consider a corresponding minimal set of U_i's with local sections $s_i : U_i \to E$ of p. Observe that $s_i|_{B'} : U_i \cap B' \to E'$ is then a local section of p', so $\operatorname{secat}(p') \leq \operatorname{secat}(p)$. If $\operatorname{secat}(p) = \infty$, we trivially conclude $\operatorname{secat}(p') \leq \infty = \operatorname{secat}(p)$, so in either case the Schwarz genus of p' is bounded above by that of p. [1] $\square$

Lemma 1.3. *For any fibration $p : E \to B$,* $\operatorname{secat}(p) \leq \operatorname{cat}(B)$.

Proof. [2] Take $\operatorname{cat}(B)$-many open sets U_i with continuous sections $s_i : U_i \to P_0B$ of $\pi_0 : P_0B \to B$. Observe that each s_i induces a homotopy $h_i : U_i \times [0, 1] \to B$ via $h_i(u, t) := s_i(u)(t)$. Then with the base point $b_0 \in B$, we can fix a corresponding element $e_0 \in p^{-1}[\{b_0\}] \subseteq E$ and define $\widetilde{h_i} : U_i \times \{0\} \to E$ by $\widetilde{h_i}(u, 0) = e_0$. This gives us that $p(\widetilde{h_i}(u, 0)) = p(e_0) = b_0 = s_i(u)(0) = h_i(u, 0)$, so by the homotopy lifting property of p there exists a homotopy $\widetilde{h_i} : U_i \times [0, 1] \to E$ such that $p \circ \widetilde{h_i} = h_i$. Then observe we can define maps $r_i : U_i \to E$ via $r_i(u) := \widetilde{h_i}(u, 1)$, giving us $p(r_i(u)) = p(\widetilde{h_i}(u, 1)) = h_i(u, 1) = u$. We therefore conclude that each r_i is a continuous section of p and so $\operatorname{secat}(p) \leq \operatorname{cat}(B)$. $\square$

Proposition 1.1. *For a path-connected space X,* $\operatorname{cat}(X) \leq \mathbf{TC}_2(X) \leq \operatorname{cat}(X \times X)$.

Proof. We have the fibration $\pi : PX \to X \times X$ from before, as well as $\pi_0 : P_0X \to X \times X$ given by $\pi_0(\gamma) = (x_0, \gamma(1))$. Observe that $\operatorname{cat}(X)$ is exactly the Schwarz genus of π_0 and $P_0X = \pi^{-1}[\{x_0\} \times X]$, so we immediately know by lemma **??** that $\operatorname{cat}(X) \leq \mathbf{TC}_2(X)$. Then by lemma 1.3 we conclude $\mathbf{TC}_2(X) = \operatorname{secat}(\pi : PX \to X \times X) \leq \operatorname{cat}(X \times X)$. $\square$

Theorem 1.1. $\mathbf{TC}_2$ *is a homotopy invariant.*

Definition 1.15. *The* order of instability *of the composition in definition 1.8 is the maximum r such that there is some sequence of indices $1 \leq i_1 < \cdots < i_r \leq k$ where $\overline{F_{i_1}} \cap \cdots \cap \overline{F_{i_r}}$ is non-empty. The* order of instability *of a motion planning algorithm s is then the minimum order of instability of decompositions for s.*

Definition 1.16. *The topological complexity $\mathbf{TC}_3(X)$ is the minimum order of instability of all tame motion planning algorithms in X.*

Remark 1.5. $\mathbf{TC}_3(X) \leq \mathbf{TC}_1(X)$.

Proof. This immediately follows from definitions 1.10 and 1.16. $\square$

[1] If one wishes to more precisely compare cardinalities, the reasoning in the finite case works.

[2] Consulted Claude for the basic idea in this proof.

Definition 1.17. *A* random n-valued path σ *in* X *is a pair of* n*-tuples* $(\gamma_1, ..., \gamma_n) \in (PX)^n$ *and* $(p_1, ..., p_n) \in [0,1]^n$ *such that all the* γ_i*'s have the same start and end points and* $p_1 + \cdots + p_n = 1$*. We write* σ *as a formal linear combination* $\sigma = p_1\gamma_1 + \cdots + p_n\gamma_n$*.*

We then define P_nX *to be the set of all* n*-valued random paths in* X *with the subspace topology induced by* $\bigoplus_n PX$*, and we have the same fibration* $\pi : P_nX \to X \times X$ *as before taking* n*-valued random paths to their end points.*

Definition 1.18. *An* n-valued random motion planning algorithm *is a continuous section* $s : X \times X \to P_nX$ *of* π*.*

Observe that $s(x, y)$ *is simply a probability distribution on* n *paths from* x *to* y*.*

Definition 1.19. *The topological complexity* $\mathbf{TC}_4(X)$ *is the minimum integer* n *such that there is some* n*-valued random motion planning algorithm* $s : X \times X \to P_nX$*.*

Theorem 1.2. *If* X *is a simplicial polyhedron, then*

$$\mathbf{TC}_1(X) = \mathbf{TC}_2(X) = \mathbf{TC}_3(X) = \mathbf{TC}_4(X).$$

Proof sketch. The proof proceeds in steps:

Step 1. $\mathbf{TC}_1(X) \leq \mathbf{TC}_2(X)$: Expand each F_i to a U_i that deformation retracts back onto it and use the deformation retract to extend $s|_{F_i} : F_i \to PX$ so $s_i : U_i \to PX$.

Step 2. $\mathbf{TC}_2(X) \leq \mathbf{TC}_1(X)$: Trim each U_i just enough to get disjoint F_i's that are ENRs.

Step 3. $\mathbf{TC}_3(X) \leq \mathbf{TC}_2(X)$: Same basic idea as step 1.

Step 4. $\mathbf{TC}_1(X) \leq \mathbf{TC}_3(X)$: Done already in remark 1.5.

Step 5. $\mathbf{TC}_2(X) \leq \mathbf{TC}_4(X)$: Construct open sets $U_j \subseteq X \times X$ from the functions $p_j : X \times X \to [0,1]$, and conclude by construction.

Step 6. $\mathbf{TC}_4(X) \leq \mathbf{TC}_2(X)$: Extend the s_i's arbitrarily, weight them via a partition of unity, and conclude.

$\square$

Definition 1.20. *Moving forward, we define the* topological complexity *of a space* X *to be* $\mathbf{TC}(X) := \mathbf{TC}_2(X)$*.*

Definition 1.21. *The* order *of an open cover* $\mathcal{U}$ *of* X *is the least integer* n *such that each point of* X *belongs to at most* n *sets of* $\mathcal{U}$ *(if such an integer exists) [Wik26c].*

Definition 1.22. *The* covering dimension *of a space* X*, denoted* $\dim X$*, to be the least integer* n *such that every finite open cover* $\mathcal{U}$ *of* X *has a refinement* $\mathcal{V}$ *of order* $n + 1$*. If no such* n *exists, then we set* $\dim X = \infty$ *[Wik26c].*

Lemma 1.4. *Let* X *be a path-connected paracompact locally contractible space. Then*

$$\operatorname{cat}(X) \leq \dim(X) + 1.$$

Lemma 1.5. *Let X be a path-connected paracompact locally contractible space. Then*

$$\dim(X \times X) = 2\dim X.$$

Theorem 1.3. *Let X be a path-connected paracompact locally contractible space. Then*

$$\mathbf{TC}(X) \leq 2\dim X + 1.$$

Proof. By lemmas 1.3, 1.4, and 1.5 we have

$$\mathbf{TC}(X) \leq \operatorname{cat}(X \times X) \leq \dim(X \times X) + 1 = 2\dim X + 1.$$

□

Theorem 1.4. *Let X be an r-connected CW-complex. Then*

$$\mathbf{TC}(X) \leq \frac{2\dim X + 1}{r+1} + 1.$$

Definition 1.23. *The tensor product $H^\bullet(X) \otimes H^\bullet(X)$ is a graded algebra with multiplication given by*

$$(u_1 \otimes v_1) \cdot (u_2 \otimes v_2) = (-1)^{|v_1||u_2|}(u_1 u_2) \otimes (v_1 v_2)$$

where $|x|$ denotes the degree of the cohomology class of x.

Definition 1.24. *The* cup product $\smile\colon H^k(X) \otimes H^\ell(X) \to H^{k+\ell}(X)$ *is given by*

$$(\alpha \smile \beta)(\sigma) := \alpha(\sigma_{[0,\ldots,k]}) \cdot \beta(\sigma_{[k,\ldots,k+\ell]}).$$

Note that this means $\smile$ induces a graded algebra $H^\bullet(X)$.

Definition 1.25. *The* ideal of the zero-divisors *of $H^\bullet(X)$ is the kernel of*

$$\smile\colon H^\bullet(X) \otimes H^\bullet(X) \to H^\bullet(X).$$

Definition 1.26. *The* zero-divisors-cup-length *of $H^\bullet(X)$ is the length of the longest non-trivial product in the ideal of zero-divisors of $H^\bullet(X)$.*

Theorem 1.5. $\mathbf{TC}(X)$ *is greater than the zero-divisors-cup-length of $H^\bullet(X)$.*

Example 1.1. *Consider S^n, letting $u \in H^n(S^n)$ be the fundamental class and $1 \in H^0(S^n)$ be the unit. Recall that $H^k(S^n) = 0$ for $k \neq 0, n$, and note $H^n(S^n) = \langle u \rangle$ and $H^0(S^n) = \langle 1 \rangle$. We then set $a = u \otimes 1 - 1 \otimes u$ and $b = u \otimes u$, and note that, up to coefficients, a and b are the only (potentially) interesting elements of $H^\bullet(S^n) \otimes H^\bullet(S^n)$. Observe that $u^2 \in H^{2n}(S^n) = 0$ so $u^2 = 0$ and we compute*

$$\begin{aligned}
a \cdot a &= (u \otimes 1 - 1 \otimes u) \cdot (u \otimes 1 - 1 \otimes u) \\
&= (u \otimes 1) \cdot (u \otimes 1) - (u \otimes 1) \cdot (1 \otimes u) - (1 \otimes u) \cdot (u \otimes 1) + (1 \otimes u) \cdot (1 \otimes u) \\
&= (-1)^{|1||u|}(u^2 \otimes 1^2) - (-1)^{|1||1|}(u \otimes u) - (-1)^{|u||u|}(u \otimes u) + (-1)^{|u||1|}(1^2 \otimes u^2) \\
&= -u \otimes u - (-1)^{n^2}(u \otimes u) \\
&= ((-1)^{n+1} - 1)u \otimes u.
\end{aligned}$$

So if n is even then $a^2 = -2b$ and if n is odd $a^2 = 0$, giving us that the zero-divisors-cup-length of $H^\bullet(S^n)$ is 1 for odd n and 2 for even n.

Theorem 1.6. *For spaces X and Y,*

$$\mathbf{TC}(X \times Y) \leq \mathbf{TC}(X) + \mathbf{TC}(Y) - 1.$$

Corollary 1.1. *For spaces $X_1, ..., X_n$ with $\mathbf{TC}(X_i) \leq M$ for all $1 \leq i \leq n$,*

$$\mathbf{TC}(X_1 \times \cdots \times X_n) \leq nM - 1.$$

Theorem 1.7. *For n path-connected spaces X_i that are (co)homologically nontrivial,*

$$\mathbf{TC}(X_1 \times \cdots \times X_n) \geq n + 1.$$

Corollary 1.2. *For n path-connected spaces X_i that are (co)homologically nontrivial,*

$$\mathbf{TC}(X_1 \times \cdots \times X_n)$$

grows linearly with n.

Proof. This follows directly from corollary 1.1 and theorem 1.7. $\square$

Proposition 1.2. *For $U \subseteq X \times X$, there exists a section $s : U \to PX$ of $\pi : PX \to X \times X$ if and only if the inclusion $i : U \hookrightarrow X \times X$ is homotopic to some diagonal map $f : U \to X \times X$ (i.e. $f[U]$ is contained in the diagonal of $X \times X$).*

Proof. First assume we have a section $s : U \to PX$ of π and consider the diagonal map $f : U \to X \times X$ given by $f(x, y) := (y, y)$. Then we can construct a homotopy $h_t : U \to X \times X$ by $h_t(x, y) := (s(x, y)(t), y)$. Observe that $h_0(x, y) = (x, y) = i(x, y)$ and $h_1(x, y) = (y, y) = f(x, y)$ so we have a homotopy from $i : U \to X \times X$ to a diagonal map $f : U \to X \times X$.

Conversely, assume we have a homotopy $h_t : U \to X \times X$ such that $h_0 = i$ and $h_1 =: f$ is a diagonal map; denote $\widehat{f} = f_1 = f_2$. Then for some $(x, y) \in U$, we have two paths, $\gamma_1(t) = (h_t(x, y))_1$ from x to $\widehat{f}(x, y)$ and $\gamma_2(t) = (h_t(x, y))_2$ from y to $\widehat{f}(x, y)$. Then we can define $s : U \to PX$ by $s(x, y) = \overline{\gamma_2} * \gamma_1$, and by construction we have $\pi \circ s = \mathrm{id}$ $\square$

Corollary 1.3. *The topological complexity $\mathbf{TC}(X) = \mathbf{TC}_2(X)$ is the smallest integer k such that $X \times X$ can be covered by k open sets where each inclusion $U_j \hookrightarrow X \times X$ is homotopic to a diagonal map $f : U \to X \times X$.*

Proof. This follows immediately from definition 1.14 and proposition 1.2. $\square$

Proposition 1.3. *If X is an ENR then*

$$\mathbf{TC}(X) \geq \mathrm{cat}((X \times X)/\Delta X) - 1$$

where $\Delta X \subseteq X \times X$ is the diagonal.

1.2 Background from [CG13]

Definition 1.27. *The* sectional category *of a map $p : E \to B$, denoted* $\mathrm{secat}(p)$*, is the least integer k such that B may be covered by k open sets U_j that each admit a map $s_j : U_j \to E$ where $p \circ s_j$ is homotopic to the inclusion $i_j : U_j \hookrightarrow B$.*

Remark 1.6. *For a fibration $p : E \to B$, definition 1.27 corresponds to definition 1.11 given in [Far06].*

Proof. First, let U_j and $s_j : U_j \to E$ be as in definition 1.11, i.e. $p \circ s_j = i_j$. Then trivially $p \circ s_i \simeq i_j$.

Conversely, let U_j and $s_j : U_j \to E$ be as in definition 1.27, i.e. $p \circ s_j \simeq i_j$, and fix some $j \leq k$. Let $h_t : U_j \to B$ be the homotopy witnessing $p \circ s_j \simeq i_j$ with $h_0 = p \circ s_j$ and $h_1 = i_j$. We define the map $\widetilde{h_0} : U_j \to E$ by $\widetilde{h_0} = s_j$, and observe that by construction $p \circ \widetilde{h_0} = p \circ s_j = h_0$. Then by the homotopy lifting property we have a homotopy $\widetilde{h_t} : U_j \to E$ such that $p \circ \widetilde{h_t} = h_t$. In particular, this gives us a map $\widehat{s_j} := \widetilde{h_1} U_j \to E$ such that $p \circ \widehat{s_j} = h_1 = i_j$. □

Definition 1.28. *The* Lusternik-Schnirelmann category *of a space X, denoted* $\mathrm{cat}(X)$*, is the least integer k such that X may be covered by k open sets U_j where each inclusion $i_j : U_j \hookrightarrow X$ is null-homotopic.*

Remark 1.7. *For a path-connected space X, definition 1.28 corresponds to definition 1.13 given in [Far06]. Note that definition 1.13 only makes sense if X is path-connected anyway.*

Proof. First, let $U_j \subseteq X$ and $s_j : U_j \to P_0X$ be as in definition 1.13, i.e. $\pi_0 \circ s_j = i_j$. We then simply define a homotopy $h_t : U_j \to X$ via $h_t(x) = s_j(x)(t)$, so $h_0(x) = s_j(x)(0) = x_0$ and $h_1(x) = s_j(x)(1) = x$. Therefore $h_0 = c_{x_0}$ and $h_1 = i_j$, so $c_{x_0} \simeq i_j$.

Conversely, let $U_j \subseteq X$ be as in definition 1.28 with $i_j \simeq c_{x_j}$ for some $x_j \in X$. Since X is path-connected all constant maps are homotopic, so without loss of generality each $x_j = x_0$. Fixing some j, let $h_t : U_j \to X$ be the homotopy witnessing $i_j \simeq c_{x_0}$ where $h_0 = c_{x_0}$ and $h_1 = i_j$. Then we define $s_j : U_j \to P_0X$ by $s_j(x)(t) = h_t(x)$, so

$$\pi_0(s_j(x)) = s_j(x)(1) = h_1(x) = i_j(x) = x.$$

□

Proposition 1.4. *For any spaces E and B along with a surjective map $p : E \to B$,* $\mathrm{secat}(p) \leq \mathrm{cat}(B)$.

Proof. Take $\mathrm{cat}(B)$-many open sets U_j such that $B = \cup_j U_j$ and each inclusion $i_j : U_j \hookrightarrow B$ is homotopic to a constant map $c_j : U_j \to B$. For each j, fix $e_j \in p^{-1}[c_j[U_j]]$ and define $s_j : U_j \to E$ by $s_j = c_{e_j}$. Observe that $p \circ s_j = c_j$ by definition, so $p \circ s_j \simeq i_j$. Therefore $\mathrm{secat}(p) \leq \mathrm{cat}(B)$. □

Proposition 1.5. *For a contractible space E and any space B along with a map $p : E \to B$,* $\mathrm{cat}(B) \leq \mathrm{secat}(p)$.

Proof. Take $\operatorname{secat}(p)$-many open sets U_j with $s_j : U_j \to E$ such that $B = \cup_j U_j$ and $p \circ s_j \simeq i_j$. Since E is contractible we have that each map s_j is homotopic to a constant map c_j, and immediately this means that $p \circ s_j$ is homotopic to $p \circ c_j$ which is also constant. Therefore $i_j \simeq p \circ c_j$ and U_j is null-homotopic so $\operatorname{cat}(B) \leq \operatorname{secat}(p)$. □

Corollary 1.4. *If E is a contractible space, B is any space, and $p : E \to B$ is a surjection, then* $\operatorname{secat}(p) = \operatorname{cat}(B)$.

Proof. This directly follows from propositions 1.4 and 1.5. □

Definition 1.29. *Given a commutative ring R and an ideal $I \subseteq R$, the* nilpotency *of I, denoted* $\operatorname{nil} I$, *is the maximum integer k such that we can find elements $r_1, ..., r_k \in I$ with $r_1 \cdots r_k \neq 0$.*

Proposition 1.6. *Let $p : E \to B$ be a fibration and let $p^* : H^*(B) \to H^*(E)$ be the induced homomorphism on cohomology. Then* $\operatorname{secat}(p) > \operatorname{nil} \ker p^*$.

Definition 1.30. *Let G be a topological group and X be a space, and assume we have some continuous G-action on X (i.e. $(g, x) \mapsto g \cdot x$ is continuous). Then X is a G-*space.

Definition 1.31. *For $x \in X$ a G-space, the* isotropy group *or* stabilizer group *of x is*

$$G_x := \{g \in G : gx = x\}.$$

Remark 1.8. *G_x is closed for all $x \in X$.*

Proof. Fix some $x \in X$ and consider a point $g \in \overline{G_x}$. Then there is a net $\{g_d\}_{d \in D} \subseteq G_x$ such that $g_d \to g$. Since each $g_d x = x$ by definition, by continuity we have that $g_d x \to gx = x$, so $g \in G_x$ and G_x is closed. □

Definition 1.32. *Given a G-space X and $x \in X$, the* orbit *of x is*

$$Gx := \{gx : g \in G\}.$$

Remark 1.9. *The map $\phi : G/G_x \to Gx$ given by $\phi(gG_x) = gx$ is a homeomorphism.*

Proof. First, let $g, g' \in G$ such that $gG_x = g'G_x$; we wish to show that $gx = g'x$. Since $gG_x = g'G_x$, there exists some $h \in G_x$ such that $ge = g'h$. Then we have

$$gx = gex = g'hx = g'x.$$

So ϕ is well-defined.

To see that ϕ is continuous, consider the quotient map $q : G \to G/G_x$; we wish to show that $\phi \circ q : G \to Gx$ is continuous. Observe that $\phi(q(g)) = \phi(gG_x) = gx$, so it immediately follows that $\phi \circ q$ is continuous.

Lastly, to see that ϕ is an open map, consider some open set $UG_x \subseteq G/G_x$. Then $\phi[UG_x] = Ux$ which is clearly open.

Therefore we conclude that ϕ is a homeomorphism. □

Definition 1.33. *For a G-space X, the* orbit space *is X/G.*

Proposition 1.7. *If G is compact and X is a Hausdorff G-space, then X/G is Hausdorff.*

Proof. [3] First consider the equivalence relation given on X where xRy if x and y lie in the same orbit. This equivalence relation is then given by the following set:

$$R := \{(x, gx) : x \in X, g \in G\} = \bigcup_{x \in X} \{x\} \times Gx \subseteq X \times X.$$

We wish to show that R is closed, so let $\{(x_a, g_a x_a)\}_{a \in A} \subseteq R$ be a convergent net with $(x_a, g_a x_a) \to (x, y) \in X \times X$. Since G is compact, the net $\{g_a\}_{a \in A}$ has a convergent subnet $\{g_b\}_{b \in B}$ with $g_b \to g \in G$. Then by continuity we have that the subnet $\{(x_b, g_b x_b)\}_{b \in B}$ converges to $(x, gx) \in R$ so $y = gx$. Thus R is closed.

Additionally, observe that $x \mapsto gx$ is a homeomorphism for any $g \in G$, so for any open $U \subseteq X$, $GU = \cup_{g \in G} gU$ is open as well. Then with the quotient map $q : X \to X/G$, $q^{-1}[GU] = GU$ since GU is a union of orbits, so $GU \in X/G$ is open.

Now let $Gx, Gy \in X/G$ with $Gx \neq Gy$, so clearly x and y do not lie in the same orbit and $(x, y) \notin R$. Because R is closed in $X \times X$, we can find open sets $U, V \subseteq X$ such that $(x, y) \in U \times V$ and $(U \times V) \cap D = \varnothing$. We immediately have that $GU \cap GV = \varnothing$ as well since otherwise x and y would be in the same orbit. Therefore we conclude that X/G is Hausdorff. $\square$

Definition 1.34. *Given two G-spaces X and Y, $f : X \to Y$ is a G-*map *if for all $g \in G$, $f(gx) = g \cdot f(x)$. We write $f : X \to_G Y$.*

Definition 1.35. *Given two G-spaces X and Y, $h : X \times [0, 1] \to Y$ is a G-*homotopy *if h is a homotopy and $h(-, t)$ is a G-map for all $t \in [0, 1]$. Equivalently, h is a G-map with G acting trivially on $[0, 1]$ and diagonally on $X \times [0, 1]$. We write $h : X \times [0, 1] \to_G Y$ or $h_t : X \to_G Y$.*

Definition 1.36. *Two G-maps $f, g : X \to_G Y$ are G-*homotopic *if there is a G-homotopy $h_t : X \to_G Y$ such that $h_0 = f$ and $h_1 = g$. We write $f \simeq_G g$.*

Definition 1.37. *Two G-spaces X and Y are G-*homotopy equivalent *if there are G-maps $f : X \to_G Y$ and $g : Y \to_G X$ such that $g \circ f \simeq_G \mathrm{id}_X$ and $f \circ g \simeq_G \mathrm{id}_Y$.*

Definition 1.38. *A set $U \subseteq X$ is* invariant *if $gU \subseteq U$ for all $g \in G$. Note that $U \subseteq GU$ always, so U is invariant if and only if $U = GU$.*

Definition 1.39. *A G-map $f : X \to Y$ is a* fixed orbit map *if $f[X]$ is contained in a single orbit in Y.*

Definition 1.40. *An invariant set $U \subseteq X$ is G-*categorical *if the inclusion $i : U \to X$ is G-homotopic to a fixed orbit map.*

Definition 1.41. *The* equivariant category *of a G-space X, denoted $\mathrm{cat}_G(X)$, is the least integer k such that X can be covered by k G-categorical open sets.*

Definition 1.42. *A G-space X is G-*contractible *if $\mathrm{cat}_G(X) = 1$.*

[3] Got proof idea from this conversation with Claude.

Proposition 1.8. *X is G-contractible if and only if X is G-homotopy equivalent to G.*

Proof. First assume that X is G-contractible, and let $h_t : X \to_G X$ be a corresponding G-homotopy witnessing this where $h_0 = \mathrm{id}_X$ and h_1 is a fixed orbit map. Let $x_0 \in X$ be some representative of the orbit $h_1[X]$, i.e. $Gx_0 = h_1[X]$. Observe that Gx_0 is trivially homeomorphic to G via the map $gx_0 \mapsto g$, so we conclude that $X \simeq_G G$.

Conversely, assume that $X \simeq_G G$ via $f : X \to_G G$ and $g : G \to_G X$ with corresponding G-homotopy $h_t : X \to_G X$ where $h_0 = g \circ f$ and $h_1 = \mathrm{id}_X$. Observe that g must be a fixed orbit map since $g(h \cdot e) = h \cdot g(e)$ so $g[G] = Gg(e)$. Then $g \circ f$ is trivially a fixed orbit map since $(g \circ f)[X] \subseteq g[G]$, so we conclude that X is G-contractible via h_t. ☐

Corollary 1.5. *For any topological group G acting on itself in the usual way, we have that $\mathrm{cat}_G(G) = 1$.*

Proof. Clearly $G \simeq_G G$, so by proposition 1.8 we are done. ☐

Proposition 1.9. *For any G-space X, $\mathrm{cat}_G(X) \geq \mathrm{cat}(X/G)$.*

Proof. Take $\mathrm{cat}_G(X)$-many G-categorical open sets U_j that cover X. Since each U_j is invariant, $U_j = GU_j \subseteq X/G$ is open. Then $i_j : U_j \hookrightarrow X$ is G-homotopic to a fixed orbit map, so immediately we have that $q \circ i_j : GU_j \hookrightarrow X/G$ is homotopic to a constant map and each GU_j is null homotopic with $X/G = \cup_j GU_j$. ☐

Proposition 1.10. *If X is metrizable and acted freely upon by G, then $\mathrm{cat}_G(X) = \mathrm{cat}(X/G)$.*

Definition 1.43. *The* equivariant sectional category *of a G-map $p : E \to_G B$, denoted $\mathrm{secat}_G(p)$, is the least integer k such that B can be covered by k invariant open sets U_j that each admit a local G-homotopy section, i.e. there exists $s_j : U_j \to E$ such that $p \circ s \simeq_G i_j$ where $i_j : U_j \hookrightarrow B$ is the inclusion.*

Proposition 1.11. *If $p : E \to_G B$ is a G-fibration, then $\mathrm{secat}_G(p) \leq k$ if and only if B can by covered by k invariant open sets that each admit a local G-section.*

Proof. This follows by an argument analogous to remark 1.6. ☐

Definition 1.44. *Given a G-space X and a closed subgroup $H \subseteq G$, the H-*fixed point set *of X is*

$$X^H := \{x \in X : \forall h \in H, hx = x\} = \{x \in X : Hx = \{x\}\}.$$

Definition 1.45. *A G-space X is G-*connected *if X^H is path-connected for every closed subgroup $H \subseteq G$.*

Proposition 1.12. *Let $p : E \to B$ be a G-map where B is G-connected and $E^G \neq \varnothing$. Then $\mathrm{secat}_G(p) \leq \mathrm{cat}_G(B)$.*

Proposition 1.13. *For a G-map $p : E \to_G B$, if $p[E^H] = B^H$ for all closed subgroups $H \subseteq G$, then $\mathrm{secat}_G(p) \leq \mathrm{cat}_G(B)$.*

Proposition 1.14. *If E is G-contractible then for any G-map $p : E \to_G B$ we have $\mathrm{cat}_G(B) \leq \mathrm{secat}_G(p)$.*

Proof. Since E is G-contractible, we have that $\mathrm{id}_E \simeq_G c$ where $c : E \to_G E$ is a fixed orbit map. Now take $\mathrm{secat}_G(p)$-many invariant open sets $U_j \subseteq B$ such that $B = \cup_j U_j$ with maps $s_j : U_j \to E$ such that $s_j \simeq_G i_j$ where $i_j : U_j \hookrightarrow B$ is the inclusion. Then we have

$$i_j \simeq_G p \circ s_j = p \circ \mathrm{id}_E \circ s_j \simeq_G p \circ c \circ s_j$$

in which the latter map is a fixed orbit map since c is a fixed orbit map. Hence we conclude $\mathrm{cat}_G(B) \leq \mathrm{secat}_G(p)$. For clarity, this is represented in the following commutative diagram:

$$\begin{tikzcd} E \arrow[r, shift left, "\mathrm{id}_E"] \arrow[r, shift right, "c"'] \arrow[dr, "p"'] & E \arrow[d, "p"] \\ U_j \arrow[u, "s_j"] \arrow[r, "i_j"'] & B \end{tikzcd}$$

□

Corollary 1.6. *Let $p : E \to_G B$ be a G-map and let E be a G-contracible space such that one of the following holds:*

1. *B is G-connected and $E^G \neq \varnothing$*
2. *$p[E^H] = B^H$ for all closed subgroups $H \subseteq G$.*

Then $\mathrm{secat}_G(p) = \mathrm{cat}_G(p)$.

Proof. This immediately follows from propositions 1.12, 1.13, and 1.14. □

1.3 Background from [DLS24]

Definition 1.46. *A map $f : X \to Y$ is* (G-)invariant *if for all $x \in X$ and for all $g \in G$, $f(g \cdot v) = f(v)$.*

Definition 1.47. *For a space X and group G, we denote X as a G-space with G acting trivially by X_{triv}.*

Definition 1.48. *We denote the category of topological spaces as* ***Top*** *and the category of G-spaces for fixed group G as G-**Top**.*

Remark 1.10. *A map $f : X \to Y$ is invariant if and only if it is a G-map when G acts trivially on Y, i.e. $f : X \to_G Y_{triv}$.*

Remark 1.11. *An invariant map $f : X \to Y$ induces a map $\widehat{f} : X/G \to Y$ via $\widehat{f}(Gv) = f(v)$. Conversely, a map $\widehat{f} : X/G \to Y$ induces an invariant map $f : X \to Y$ via $f(v) = \widehat{f}(Gv)$. This gives an isomorphism $\boldsymbol{Top}(X/G, Y) \cong G\text{-}\boldsymbol{Top}(X, Y_{triv})$.*

Furthermore, let $L : G\text{-}\boldsymbol{Top} \to \boldsymbol{Top}$ and $R : \boldsymbol{Top} \to G\text{-}\boldsymbol{Top}$ be the functors defined as follows:

- $LX = X/G$, $L(f : X \to_G Y) = (Gx \mapsto G(f(x)))$
- $RY = Y_{triv}$, $R(f : Y \to X) = (y \mapsto f(y))$.

Then the previous isomorphism actually means that L is a left adjoint of R.

Definition 1.49. *Given a G-space X, an* orbit canonicalization *is an invariant map $f : X \to X$ such that $f(x) \in Gx$ for all $x \in X$.*

Remark 1.12. *An orbit canonicalization $f : X \to X$ must have a unique fixed point in each orbit $Gx \in X/G$.*

Proof. Remark 1.11 gives us an embedding $X/G \hookrightarrow X$, and we conclude by instead considering this embedding as a map $X \to X$. ☐

Proposition 1.15. *Every orbit canonicalization $f : X \to X$ corresponds uniquely to a continuous section $s : X/G \to X$.*

Proof. First, let $f : X \to X$ be an orbit canonicalization. As per remark 1.11, f induces $\widehat{f} : X/G \to X$ via $\widehat{f}(Gx) = f(x)$ with $\widehat{f} \circ q = f$. Since f is invariant, we have that $q(\widehat{f}(Gx)) = q(f(x)) = G(f(x)) = Gx$ and so $\widehat{f}$ is a section of q.

Conversely, let $s : X/G \to X$ be some section of q such that $s(Gx) \in Gx$ always. Then we can define a map $f : X \to X$ via $f = s \circ q$ and observe that $f(x) = s(q(x)) = s(Gx)$. Since s is a section of q, we have that $q(s(Gx)) = Gx$, meaning that $f(x) = s(Gx) \in q^{-1}[Gx] = Gx$. Therefore f is an orbit canonicalization.

Lastly, to see the uniqueness, observe that the process in the converse step applied to $\widehat{f}$ yields f, so the map $f \mapsto \widehat{f}$ is a bijection. ☐

Corollary 1.7. *An orbit canonicalization $f : X \to X$ exists if and only if there is a continuous section $s : X/G \to X$.*

Proof. This is just a restatement of proposition 1.15. ☐

Chapter 2

New Work

2.1 Sectional Category and Canonicalization

Proposition 2.1. *If there exists an orbit canonicalization $f : X \to X$, then* $\text{secat}(q) = 1$ *where $q : X \to X/G$ is the quotient map.*

Proof. By proposition 1.15 we know that f induces a section $\widehat{f} : X/G \to X$ of q, so we conclude that $\text{secat}(q) = 1$. $\square$

Corollary 2.1. *If* $\text{secat}(q) > 1$*, there is no orbit canonicalization $f : X \to X$.*

With this in mind, we now wish to study the sectional category of the quotient map $q : X \to X/G$.

Definition 2.1. *For spaces E and B, a map $p : E \to B$ has the* homotopy lifting property up to homotopy (HLPH) *for a space X if for every homotopy $h_t : X \to B$ and every homotopy lift $\widetilde{h_0} : X \to E$ of h_0 ($h_0 \simeq p \circ \widetilde{h_0}$), there exists a homotopy $\widetilde{h_t} : X \to E$ that lifts h_t up to homotopy ($h_t \simeq p \circ \widetilde{h_t}$).*

Proposition 2.2. *For any spaces E and B and a surjective map $p : E \to B$,* $\text{secat}(p) \leq \text{cat}(B)$ *as in definitions 1.11 and 1.13.*

Proof. Take $\text{cat}(B)$-many open sets U_j such that $B = \cup_j U_j$ and each inclusion $i_j : U_j \hookrightarrow B$ is homotopic to a constant map $c_j : U_j \to B$. $\square$

Definition 2.2. *For spaces E and B, a map $p : E \to B$ is a* fibration up to homotopy *if p satisfies the HLPH for all topological spaces.*

Remark 2.1. *Clearly HLP implies HLPH and any fibration is a fibration up to homotopy.*

Lemma 2.1. *Let X be a G-space and let $q : X \to X/G$ be the quotient map. Then if $U \subseteq X$ is G-categorical, the set $q[U] \subseteq X/G$ is contractible.*

Proof. First, let $U \subseteq X$ be G-categorical, i.e. the inclusion $i : U \hookrightarrow X$ is G-homotopic to a fixed orbit map $c : U \to X$. This clearly induces a homotopy from the inclusion $q \circ i : q[U] \hookrightarrow X$ to $q \circ c : q[U] \to X/G$, and since c is a fixed orbit map $q \circ c$ must be a constant map. Thus $q[U]$ is contractible. $\square$

Lemma 2.2. *Let X be a G space such that the quotient map $q : X \to X/G$ is a fibration. Then if $U \subseteq X/G$ is open and contractible, the set $q^{-1}[U] \subseteq X$ is G-categorical.*

Proof. Let $h_t : U \to X/G$ be a homotopy witnessing that U is contractible, with h_0 being the inclusion $i : U \hookrightarrow X/G$ and h_1 being a constant map $c : U \to X/G$. Set $V = q^{-1}[U]$, noting that V is open and invariant, and let $\widetilde{h_0} : V \to X$ be the inclusion $\widetilde{i} : V \hookrightarrow X$. Then clearly $q \circ \widetilde{h_0} = i$, so by the homotopy lifting property we have a homotopy $\widetilde{h_t} : V \to X$ such that $q \circ \widetilde{h_t} = h_t$. Specifically this means $q \circ \widetilde{h_1} = c$, i.e. $\widetilde{h_1} : V \to X$ is a fixed orbit map, so we conclude that V is G-categorical. $\square$

Proposition 2.3. *If X is a G-space such that the quotient map $q : X \to X/G$ is a fibration up to homotopy, then* $\text{secat}(q) \leq \text{cat}_G(X)$.

Proof. Take $\text{cat}_G(X)$-many G-categorical sets $U_j \subseteq X$ such that $X = \cup_j U_j$. Define $V_j = q[U_j]$, and note that by lemma 2.1 each V_j is contractible. Fix some j and let $h_t : V_j \to X/G$ be a homotopy witnessing the contractibility of V_j, with h_0 being a constant map $c_j : V_j \to X/G$ and h_1 being the inclusion $i_j : V_j \hookrightarrow X/G$. Then fix some $x_0 \in q^{-1}[c_j[V_j]]$ and define $\widetilde{h}_0 : V_j \to X$ by $\widetilde{h}_0 = c_{x_0}$. Clearly $q \circ \widetilde{h}_0 = c_j$, so the HLPH gives us a homotopy $\widetilde{h}_t : V_j \to X$ such that $q \circ \widetilde{h}_t \simeq h_t$. In particular, this means $q \circ \widetilde{h}_1 \simeq i_j$ so $\widetilde{h}_1$ is a homotopy section of q. Repeating this for all j, we conclude that $\text{secat}(q) \leq \text{cat}_G(X)$. □

Definition 2.3. *The* contractible sectional category *of a map $p : E \to B$, denoted by* $\text{consecat}(p)$*, is the least integer k such that B may be covered by k contractible open sets U_j that each admit a map $s_j : U_j \to E$ where $p \circ s_j$ is homotopic to the inclusion $i_j : U_j \hookrightarrow B$.*

Remark 2.2. *Trivially,* $\text{consecat}(p) \geq \text{secat}(p)$ *for any map p.*

Proposition 2.4. *If X is a G-space such that the quotient map $q : X \to X/G$ is a fibration up to homotopy, then* $\text{cat}_G(X) \leq \text{consecat}(q)$.

Proof. Take $\text{consecat}(q)$-many contractible open sets $U_j \subseteq X/G$ with $X/G = \cup_j U_j$ that each admit a homotopy section $s_j : U_j \to X$ of q. Define $V_j = q^{-1}[U_j]$, and note that by lemma 2.2 that each V_j is G-categorical. Clearly $X = \cup_j V_j$, so we conclude that $\text{cat}_G(X) \leq \text{consecat}(q)$. □

Question 2.1. *The homotopy sections in the proof above are completely unused. Can something stronger be proven?*

Question 2.2. *When do we have* $\text{consecat}(p) = \text{secat}(p)$*? Or are there any other bounds we can put on* $\text{secat}(p)$ *via* $\text{consecat}(p)$*?*

Recall that when $p : E \to B$ is a fibration we have that definitions 1.11 and 1.27 coincide (i.e. homotopy sections induce sections), so we stand to learn a lot about orbit canonicalizations if we can discern when $q : X \to X/G$ is a fibration.

Theorem 2.1. *This is part of theorem 1.8 in [Rai06]. A fiber bundle $p : E \to B$ over paracompact space B is a fibration.*

2.2 Bounds for Strong and Weak Sectional Category

We first introduce some (slightly) new terminology.

Definition 2.4. *The* weak sectional category *of a map $p : E \to B$, denoted* $\mathrm{secat}(p)$*, is the least integer k such that there k open sets U_j covering B that each admit a homotopy section $s_j : U_j \to E$ of p (i.e. $p \circ s_j \simeq i_j$ where $i_j : U_j \hookrightarrow B$ is the inclusion).*

Definition 2.5. *The* strong sectional category *of a map $p : E \to B$, denoted* $\mathrm{secat}^{\mathrm{str}}(p)$*, is the least integer k such that there k open sets U_j covering B that each admit a section $s_j : U_j \to E$ of p (i.e. $p \circ s_j = i_j$ where $i_j : U_j \hookrightarrow B$ is the inclusion).*

Recall that by remark 1.6 $\mathrm{secat}(p) = \mathrm{secat}^{\mathrm{str}}(p)$ when p is a fibration. In general, we have the following result:

Lemma 2.3. *For any map $p : E \to B$,* $\mathrm{secat}(p) \leq \mathrm{secat}^{\mathrm{str}}(p)$*.*

Proof. Take $\mathrm{secat}^{\mathrm{str}}(p)$-many open sets covering U_j with sections $s_j : U_j \to E$ of p i.e. $p \circ s_j = i_j$. It trivially follows that $p \circ s_j \simeq i_j$, so we are done. □

Definition 2.6. *(Definition 1.1 in [Coh26]) Let E, B, and F be spaces, and let $p : E \to B$ be a map such that*

1. *p is surjective,*
2. *for all $b \in B$, $p^{-1}[\{b\}] \cong F$,*
3. *and for all $b \in B$, there is some open neighborhood $U_b \subseteq B$ of B and a homeomorphism $\Phi_b : p^{-1}[U_b] \to U_b \times F$ such that $p|_{p^{-1}[U_b]} = \pi_b \circ \Phi_b$ where $\pi_b : U_b \times F \to U_b$ is the projection.*

Then p is a fiber bundle with fiber F*.*

Lemma 2.4. *(Corollary 2.7.14 in [Spa66]) If B is paracompact, then any fiber bundle $p : E \to B$ is a fibration.*

Definition 2.7. *(Definition 1.5 in [Coh26]) A topological group G is a* Lie group *is G is a differentiable manifold such that products and inverses are differentiable maps.*

Definition 2.8. *(Definition 1.6 in [Coh26]) Let G be a topological group, let E and B be spaces, and let $p : E \to B$ be a fiber bundle with fiber G such that*

1. *E has a free left G-action where $p(e) = p(ge)$ for all $g \in G$, $e \in E$, i.e. the fibers of p are closed under the G-action,*
2. *for all $b \in B$, the induced G-action on $p^{-1}[\{b\}]$ is free and transitive,*
3. *and for every open set $U \subseteq B$, there exists a G-equivariant homeomorphism $\Psi_U : p^{-1}[U] \to G \times U$ where G acts on $G \times U$ by multiplication on G.*

Then p is a principal G-bundle*.*

Definition 2.9. *(Definition 1.7 in [Coh26]) A G-space X* has slices *if the quotient map $q : X \to X/G$ has local sections around every point in X/G.*

Proposition 2.5. *(Proposition 1.2 in [Coh26]) If X is a free G-space with slices, then the quotient map $q : X \to X/G$ is a principal G-bundle.*

Theorem 2.2. *(Theorem 1.3 in [Coh26]) Let M be a smooth manifold with a smooth, free G-action by a compact Lie group G. Then M has slices.*

Corollary 2.2. *Let M be a smooth manifold with a smooth, free G-action by a compact Lie group G. Then $q : M \to M/G$ is a principal G-bundle.*

Proof. This follows directly from theorem 2.2 and proposition 2.5. $\square$

Theorem 2.3. *Let M be a smooth manifold with a smooth, free G-action by a compact Lie group G such that M/G is paracompact. Then $q : M \to M/G$ is a fibration.*

Proof. This follows directly from corollary 2.2 and lemma 2.4. $\square$

Theorem 2.4. *Let G be a compact Lie group and let M be a smooth manifold with submanifold $N \subseteq M$ such that*

1. *G acts smoothly on M,*
2. *N is G-invariant (hence $N/G \subseteq M/G$),*
3. *the induced G-action on N is free,*
4. *and N/G is paracompact.*

Then letting $q : M \to M/G$ and $p = q|_N : N \to N/G$ be the respective quotient maps and letting $p^* : H^*(N/G) \to H^*(N)$ be the induced map on cohomology, we have the following chain of inequalities:

$$\operatorname{nil}\ker(p^*) < \operatorname{secat}(p) = \operatorname{secat}^{\text{str}}(p) \leq \operatorname{secat}^{\text{str}}(q).$$

Proof. Since G is a compact Lie group acting freely on N and N/G is paracompact, by theorem 2.3 we know that p is a fibration. Then by proposition 1.6 we have that $\operatorname{nil}\ker(p^*) < \operatorname{secat}(p)$; by remark 1.6 we have that $\operatorname{secat}(p) = \operatorname{secat}^{\text{str}}(p)$; and by lemma 1.2 we have that $\operatorname{secat}^{\text{str}}(p) \leq \operatorname{secat}^{\text{str}}(q)$. $\square$

Theorem 2.4 gives a concrete method to find a lower bound for strong sectional category even for quotient maps that fail to be fibrations by considering appropriate submanifolds of a given space. This is particularly useful in the problem of orbit canonicalization, since orbit canonicalizations are directly related to the strong sectional category of the quotient map.

2.3 Canonicalizing $\mathbb{R}^{d\times n}$

Definition 2.10. *The* orthogonal group *of degree n is the set*

$$\mathrm{O}(n) := \left\{Q \in \mathbb{R}^{n\times n} : QQ^T = I\right\}.$$

Definition 2.11. *The* special orthogonal group *of degree n is the set*

$$\mathrm{SO}(n) := \left\{Q \in \mathbb{R}^{n\times n} : \det(Q) = 1\right\}.$$

Remark 2.3. *Observe that* $\mathrm{SO}(n) \subseteq \mathrm{O}(n)$.

Definition 2.12. *The set of* positive semi-definite matrices of degree n *is*

$$\mathbb{S}^n_+ := \left\{A \in \mathbb{R}^{n\times n} : A = A^T, \forall x \in \mathbb{R}^n, x^T A x \geq 0\right\}.$$

Definition 2.13. *The set of* positive semi-definite matrices of degree n with at most rank d *is*

$$\mathbb{S}^n_{\leq d} := \left\{A \in \mathbb{S}^n_+ : \mathrm{rank}(A) \leq d\right\}.$$

Remark 2.4. $\mathbb{S}^n_{\leq d} = \mathbb{S}^n_{\leq n}$ *for any* $d \geq n$.

Proposition 2.6. $A \in \mathbb{S}^n_+$ *if and only if there is some matrix* $B \in \mathbb{R}^{n\times n}$ *such that* $A = B^T B$ *[Wik26a].*

Corollary 2.3. $\mathbb{S}^n_+ = \mathbb{S}^n_{\leq n}$.

Definition 2.14. *The set of* positive semi-definite matrices of degree n with rank d *is*

$$\mathbb{S}^n_d := \left\{A \in \mathbb{S}^n_+ : \mathrm{rank}(A) = d\right\}.$$

Proposition 2.7. $A \in \mathbb{S}^n_d$ *if and only if there is some matrix* $B \in \mathbb{R}^{d\times n}$ *such that* $A = B^T B$ *with* $\mathrm{rank}(B) = d$. *[Wik26a].*

Lemma 2.5. *With $O(d)$ acting on $\mathbb{R}^{d\times n}$ via matrix multiplication on the left, $\mathbb{R}^{d\times n}/\,\mathrm{O}(d)$ is homeomorphic to the space* $S := \left\{A^T A : A \in \mathbb{R}^{d\times n}\right\}$.

Proof. To begin, consider the map $\phi : \mathbb{R}^{d\times n}$ given by $\phi(A) = A^T A$. Now let $A, B \in \mathbb{R}^{d\times n}$ such that $[A] = [B] \in \mathbb{R}^{d\times n}/O(d)$, i.e. there exists $Q \in O(d)$ such that $A = QB$. This means that $A^T A = (QB)^T QB = B^T Q^T QB = B^T B$, so ϕ induces a map $\widehat{\phi} : \mathbb{R}^{d\times n}/O(d) \to S$ via $\widehat{\phi}([A]) = A^T A$. Note that $\widehat{\phi}$ is bijective by construction, with inverse $\widehat{\phi}^{-1} : S \to \mathbb{R}^{d\times n}/O(d)$ given by $\widehat{\phi}^{-1}(A^T A) = [A]$. Both of these maps are continuous, so we are done. $\square$

Corollary 2.4. $\mathbb{R}^{d\times n}/O(d) \cong \mathbb{S}^n_{\leq d}$.

Proof. This follows directly from proposition 2.7 and lemma 2.5. $\square$

With corollary 2.4 in hand, we have that any invariant map $f : \mathbb{R}^{d\times n} \to Y$ factors through $\mathbb{S}^n_d$ and so it suffices to reduce to consider maps from $\mathbb{S}^n_d$ to Y.

Definition 2.15. *Given a group G and a G-set X, the* free locus *of X is*

$$X_{free} := \{x \in X : \forall g \in G \setminus \{e\}, gx \neq x\} = \{x \in X : G_x = \{e\}\}$$

Lemma 2.6. *With $O(d)$ acting on $\mathbb{R}^{d\times n}$ via matrix multiplication, $\mathbb{R}^{d\times n}_{free} = \mathbb{R}^{d\times n}_*$.*

Proof. First, let $A \in \mathbb{R}^{d\times n}$ such that $\text{rank}(A) = k < d$. Then there are some orthonormal bases $\{e_1, ..., e_k\}$ of $\text{col}(A)$ and $\{e_{k+1}, ..., e_d\}$ of $\text{null}(A)$. Let $B = [e_1 \cdots e_d]^T \in O(d)$ be the change of basis matrix from the standard basis for $\mathbb{R}^d$ to $\{e_1, ..., e_d\}$. Then A can be written in the basis $\{e_1, ..., e_d\}$ as a matrix

$$\widehat{A} = BA = \begin{bmatrix} \widehat{a}_{1,1} & \cdots & \widehat{a}_{n,1} \\ \vdots & & \vdots \\ \widehat{a}_{k,1} & \cdots & \widehat{a}_{n,k} \\ 0 & \cdots & 0 \\ \vdots & & \vdots \\ 0 & \cdots & 0 \end{bmatrix}.$$

Now define the matrix $R \in \mathbb{R}^{d\times d}$ by $R_{i,i} = 1$ if $i \leq k$, $R_{i,i} = -1$ if $i > k$, and $R_{i,j} = 0$ if $i \neq j$ (R simply flips the sign of the last $d-k$ coordinates) and observe that clearly $R \in O(d)$. Define $Q = B^T RB$, and then $Q \in O(d)$ since

$$Q^T Q = (B^T RB)^T B^T RB = B^T R^T BB^T RB = I.$$

By construction we have that $R\widehat{A} = \widehat{A}$, so it follows that $QA = A$. However, without loss of generality $Q \neq I$, so we have that $A \notin \mathbb{R}^{d\times n}_{free}$.

Conversely, let $A \in \mathbb{R}^{d\times n}_*$ and let $Q \in O(d)$ such that $QA = A$. Since $\text{col}(A) = \mathbb{R}^d$, this means that Q fixes all of $\mathbb{R}^d$ and hence $Q = I$. Therefore $A \in \mathbb{R}^{d\times n}_{free}$. □

Corollary 2.5. *$O(d)$ does not act freely on any subset of $\mathbb{R}^{d\times n}$ if $n < d$.*

Corollary 2.6. $\mathbb{R}^{d\times n}_*/O(d) \cong \mathbb{S}^n_d$.

Proof. This follows from proposition 2.7, lemma 2.5, and lemma 2.6. □

Definition 2.16. *(From [BZA24]) The* Stiefel manifold *$V_k(n)$ is the set of orthonormal basis matrices for k-dimensional subspaces of $\mathbb{R}^n$, which can be represented via*

$$V_k(n) := \left\{U \in \mathbb{R}^{n\times k} : U^T U = I \in \mathbb{R}^{k\times k}\right\}$$

Definition 2.17. *(From [BZA24]) The* Grassmann manifold *$G_k(n)$ is the set of all k-dimensional subspaces of $\mathbb{R}^n$. This can be represented with the set of all rank k orthogonal projection matrices via*

$$G_k(n) := \left\{P \in \mathbb{R}^{n\times n} : P^T = P, P^2 = P, \text{rank}(P) = k\right\}.$$

Remark 2.5. *The Grassmann manifold $G_k(n)$ can also be realized as a quotient of the Stiefel manifold $V_k(n)$ by relating matrices that are basis for the same subspace, i.e. those that differ by an orthogonal transformation. This is captured by the action of $O(k)$ on $V_k(n)$ by right multiplication, hence $G_k(n) \cong V_k(n)/O(k)$.*

Both of these manifolds are compact, and have corresponding non-compact variants:

Definition 2.18. *The* non-compact Stiefel manifold $V'_k(n)$ *is the set of basis matrices for k-dimensional subspaces of $\mathbb{R}^n$, which can be represented via*

$$V_k(n) := \left\{U \in \mathbb{R}^{n\times k} : \operatorname{rank}(U) = k\right\}$$

Definition 2.19. *The* non-compact Grassmann manifold $G'_k(n)$ *is $V'_k(n)/O(k)$ where $O(k)$ acts on $V'_k(n)$ on the right via matrix multiplication.*

Remark 2.6. *The transposition map $A \mapsto A^T$ gives a homeomorphism between $\mathbb{R}^{d\times n}_*$ and $V'_d(n)$, hence $\mathbb{R}^{d\times n}_*/O(d) \cong G'_d(n)$.*

We now present a somewhat simplified proof of the result of [CG13] that $(\mathbb{R}^{d\times n}, O(d))$ and $(\mathbb{R}^{d\times n}, SO(d))$ lack continuous canonicalizations for $n > d$ and $n \geq d$ respectively.

Definition 2.20. *The* shape space Σ^k_m *is*

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